MILITARY SPECIFICATION

HEAD ASSEMBLY, TANK PERISCOPE: 10513625

1. SCOPE

1.1 This specification covers one type periscope head assembly which deflects a forward line of sight to the body of a tank periscope (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army Munitions Command

F10513625	Head Assembly

PACKAGING DATA SHEET

MIL-P-14232/P10513625	Packaging of Head Assembly: 10513625

FSC 6650

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(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The head assembly shall be manufactured in accordance with Drawing F10513625 and drawings pertaining thereto.

3.2 General specification.- The contractor shall be responsible for adherence to and compliance with the requirements of MIL-F-13926 specified below.

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Workmanship.

3.3 Environmental.

3.3.1 Storage temperatures.- The head assembly shall show no evidence of physical failure such as glass breakage, bonding failure, or casting failure due to storage at temperatures ranging from -80° to +160° Fahrenheit (F).

3.3.2 Vibration.- The head assembly shall show no evidence of physical failure due to being vibrated at 30 cycles per second and an amplitude of 0.06 inch (0.12 inch total excursion).

3.4 Sealing.- The head assembly shall show no evidence of water penetration between the prism and casting when exposed to prolonged rainfall. This requirement shall be met with the mounting pads in the bottom position and the cavity sealed.

3.5 Light deviation.- With the prism in the top position and the bottom surfaces of the four pads in a horizontal plane, an incident line of sight 2.0 degrees below a horizontal plane and perpendicular to an axis defined by the centers of the two 120 degree countersinks shown on Drawing F10513626 shall emerge perpendicular to the horizontal plane within 45 minutes of arc.

3.6 Image tilt.- The image of a target line originally parallel to an axis through the centers of the two countersinks shall emerge parallel to the same axis within 45 minutes of arc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable

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to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection of components and sub-assemblies not covered by specifications or Supplementary Quality Assurance Provisions (SQAP) shall be in accordance with 4.2.2. Definitions of inspection terms not otherwise defined herein shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Inspection lot.- Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examination and tests.- Components and subassemblies shall be inspected in accordance with the inspection provisions contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with MIL-STD-105, and the inspection level and sampling plans specified in Table I titled "Classification of Defects". Examinations and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II of Table I with
Sampling Plan Table II-A of MIL-STD-105

	REQUIREMENT	TEST PROCEDURES
CRITICAL: NONE		
MAJOR: AQL 1.0%		
101. Vibration	3.3.2	4.5.2.2
102. Light deviation	3.5	4.5.4
103. Image tilt	3.6	4.5.5
MINOR: NONE		

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4.2.3 Disposition of non-conforming product.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3 Control tests

4.3.1 General.- One head assembly shall be selected at random as a control sample from each 100 produced, or from each month's production, whichever occurs first. The control sample shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301. Fabrication	3.1	4.5.1
302. General specification	3.2	Visual-per MIL-F-13926

4.3.2 Environmental.- Three head assemblies shall be selected at random as a control sample from each 50 produced, or from each month's production, whichever occurs first. The control sample shall have successfully met all other acceptance tests specified herein, and shall then meet the following requirements and tests:

CONTROL TEST	REQUIREMENT	TEST PROCEDURES
303. Storage temperature	3.3.1	4.5.2.1
304. Sealing	3.4	4.5.3

4.3.3 Control test failure.- Should any one item of a control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall maintain all required measuring and testing equipment in accordance with the applicable requirements specified in MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.4.2 Contractor furnished inspection equipment.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" requirements of MIL-F-13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of product which exceeds the prescribed tolerance

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limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.3 Devices used for environmental testing.- All environmental testing devices used to determine conformance of the environmental requirements listed below shall conform to the "Test Facilities" requirements outlined in MIL-F-13926.

(a) Temperature (3.3.1)
(b) Vibration (3.3.2)
(c) Sealing (3.4)

4.5 Test methods and procedures.

4.5.1 Fabrication.- The head assembly shall be subjected to a visual, tactile and dimensional inspection to determine compliance with 3.1 and the applicable drawings and requirements of this specification. Dimensional inspection shall be performed using standard measuring equipment.

4.5.2 Environmental.

4.5.2.1 Storage temperatures.- This test shall be conducted in accordance with the temperature conditions indicated in 3.3.1. The temperatures shall be changed gradually to the specified temperature extremes and maintained at these extremes until thermal stabilization has been reached. Upon completion of the specified exposure and return to standard ambient temperature (60° to 90°F), the head assembly shall be visually inspected for evidence of physical failure such as casting damage or glass breakage. It shall then be subjected to the tests specified in 4.5.2.2 to 4.5.5 inclusive to determine conformance with the requirements referenced therein.

4.5.2.2 Vibration.- The head assembly shall be clamped to a vibration table with the prism in the top position and the bottom surfaces of the four mounting pads horizontal and parallel to the table. The head assembly shall be vibrated in a vertical plane at the amplitude and frequency specified in 3.3.2 for a period of 5 minutes ± 15 seconds. The vibration amplitude shall be measured at the upper portion of the head assembly as positioned on the vibration table. Subsequent to vibration, the head assembly shall be subjected to a visual and tactile examination and shall show no evidence of physical damage or failure. The head assembly shall then be subjected to the tests specified in 4.5.4 and 4.5.5 to determine compliance with the requirements referenced therein.

4.5.3 Sealing.- This test shall be accomplished by sealing the bottom cavity of the head assembly to prevent water penetration in the cavity during the rain test. Position the head assembly in the test chamber in an upright position with the sealed cavity on the bottom to prevent the simulated rainfall from being directed against this sealed area. Sealing shall be inspected in accordance with Procedure I of the rain test specified in MIL-F-13926. Upon completion of the test, the head

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assembly shall be removed from the test chamber and all surface water quickly absorbed with a dry cloth or blotting paper. The cavity seal shall then be opened and the area within the cavity visually examined for evidence of water penetration or damage. Visually inspect the prism surfaces for the presence of water between the prism and outer casting. There shall be no water present when determining compliance with 3.4.

4.5.4 Light deviation. -

4.5.4.1 Orientation. - This test shall be performed by utilizing a holding device to receive the head assembly on the stage of a spectrometer. The positioning of the head assembly shall be in accordance with 3.5 except that the bottom surfaces of the 4 pads shall be in a vertical plane. The axis through the two 120° countersinks shall be perpendicular to the incident line of sight. The incident line of sight shall also be 88 degrees from a plane perpendicular to the mounting pads. The viewing telescope of the spectrometer shall be provided with a horizontal and vertical reticle line, the intersection of which shall be on the optical axis. Two additional reticle lines for comparing image tilt with respect to the vertical target line representing the incident line of sight shall also be provided. The collimator of the spectrometer shall be provided with a reticle containing three vertical lines and a horizontal line. The intersecting point of the central vertical line and horizontal line shall be placed on the optical axis of the collimating telescope. The outer vertical lines are 2 degrees each side of the optical axis of the collimating telescope to permit obtaining 88 degrees from the collimating telescope to either side of the viewing telescope.

4.5.4.2 Alignment of viewing telescope to collimator. - Alignment of the optical lines of sight shall be accomplished by setting the optical axis of the viewing telescope parallel and in coincidence with the optical axis of collimator projector. The optical axis of both optical instruments shall be considered as the point of intersection between their respective horizontal and center vertical reticle lines. Record the elevation and azimuth reading on their respective knobs on the spectrometer.

4.5.4.3 Performing the actual test. - The test is performed by positioning the head assembly in the holding device on the stage as prescribed above. Position the viewing telescope at the 88 degree angle specified in 4.5.4.1 and observe coincidence between the horizontal line and respective vertical line representing this angular measurement. They must coincide within the minutes of arc specified in 3.5. All angular measurements shall be measured directly from the elevation and azimuth micrometer scales on the spectrometer.

4.5.5 Image tilt. - This test can be accomplished using the set up specified in 4.5.4.1 and observing parallelism between reticle line of the collimator representing the incident line of sight (88 degrees) and the vertical line of sight of the viewing telescope. They must remain parallel within the limits of the angular lines of the viewing telescope's reticle to conform to the requirements of 3.6.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with MIL-P-14232 and MIL-P-14232/P10513625. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Tank Periscope Head Assembly 10513625 is primarily intended for use as a component of the driver's periscopes in the XM551 Vehicle. The head assembly forms the unity power, daylight, XM47 Tank Periscope when assembled to Tank Periscope Body Assembly 10513620. This unit also forms the unity power, infrared, XM48 Tank Periscope when assembled to Tank Periscope Body Assembly 10513599.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet number (see 5.1).

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